## 1.8.5 Wrap-Up: Cool Down

1. Create an example of the Segment Addition Postulate and show the steps to finding the total length.

# Unit 1, Lesson 9: Copying an Angle

## Benchmark

MA.912.GR.5.1 Construct a copy of a segment or an angle.

## Learning Targets

- ☐ I can construct a copy of an angle.
- ☐ I can use the angle addition postulate.

## 1.9.1 Warm-Up: Bell Work

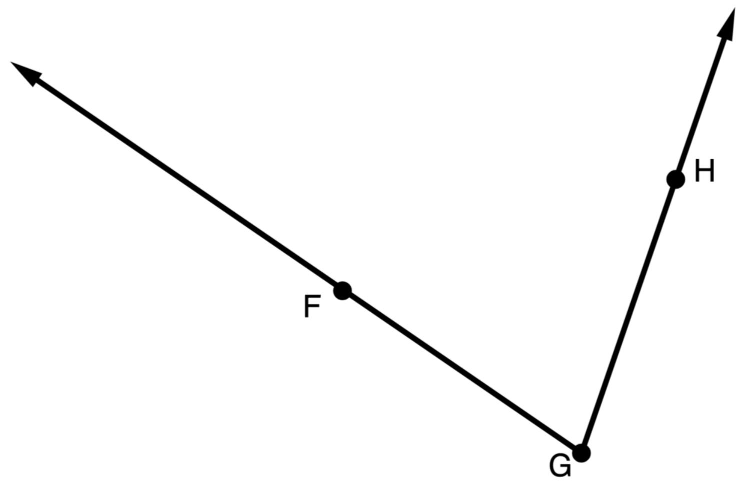
1. Points $C$, $H$, and $W$ are collinear. Point $W$ is between $C$ and $H$. $CW = 4x - 2$, $HW = 5x - 19$, and $CH = 7x + 5$.

   a. Find the value of $x$.

   b. What is the length of $\overline{HW}$?

## 1.9.2 Activity: Copying an Angle

1. $\angle FGH$ is shown.




   a. Discuss with your partner how to copy $\angle FGH$ using only a compass.

b. Work with your partner to order the nine steps for copying $\angle FGH$. The first and last step are in the correct position.

| | |
|---|---|
| 1 | Draw a point and label it $X$. Then, draw a ray starting at point $X$ in any direction. Label the ray $\overrightarrow{XZ}$. |
| | Set the compass' point on point $A$ and adjust the compass' width to point $B$. |
| | Without changing the compass' width, place the compass' point on $Y$ and draw a similar arc to create point $W$ where the two arcs meet. |
| | Set the compass' point on point $G$ and adjust the compass' width to any width. |
| | Without changing the compass' width, place the compass' point on $X$ and draw a similar arc. Label the point where the arc and $\overrightarrow{XZ}$ intersect point $Y$. |
| | Label the point where the arc crosses $\overrightarrow{GF}$ as point $A$. |
| | Label the point where the arc crosses $\overrightarrow{GH}$ as point $B$. |
| | Draw an arc across both rays of $\angle FGH$. |
| 9 | Draw $\overrightarrow{XW}$ using a straightedge. |

c. Compare your ordering with another group. If necessary, make adjustments to the order.

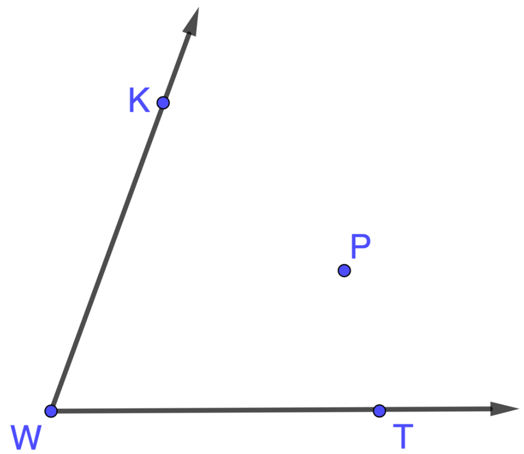
d. Use your steps to draw $\angle WXY$, a copy of $\angle FGH$. If necessary, make adjustments to the order of your steps.

## 1.9.3 Guided Instruction: Angle Addition Postulate

### Angle Addition Postulate

If point $X$ is in the interior of $\angle AMB$, then $m\angle AMX + m\angle XMB = m\angle AMB$.

1. Point $P$ lies in the interior of $\angle KWT$ as shown.




a. Draw $\overrightarrow{WP}$.

b. Use a protractor to measure each angle in degrees.

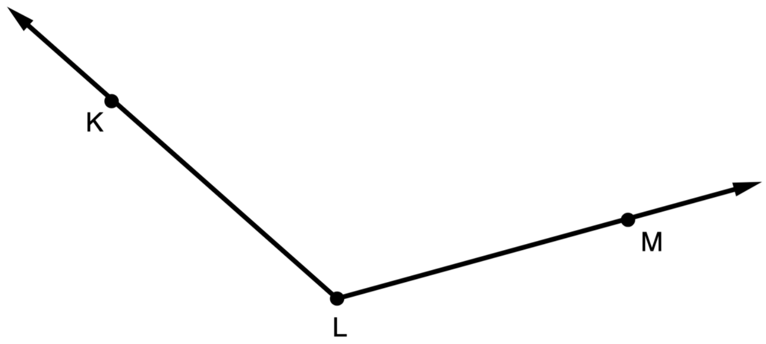
$m\angle KWT$ = ______ $m\angle KWP$ = ______ $m\angle PWT$ = ______

c. Discuss with your partner why $m\angle KWP + m\angle PWT = m\angle KWT$ should be true. Summarize your discussion.

d. If $m\angle KWT = 70°$, $m\angle KWP = (5x + 5)°$ and $m\angle PWT = (4x - 7)°$, determine the value of $x$.

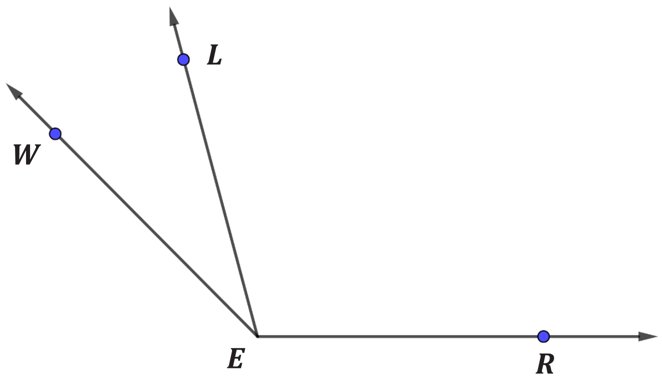
## 1.9.4 Try It: Copying an Angle

1. $\angle KLM$ is shown.




Draw $\angle RST$, a copy of $\angle KLM$.

2. Three angles are shown, $\angle WEL$, $\angle LER$, and $\angle WER$.




If $m\angle LER = (14.35x - 9.8)°$, $m\angle WEL = (2.5x + 10)°$, and $m\angle WER = 135°$, what is $m\angle WEL$?

## 1.9.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about copying an angle.

# Unit 1, Lesson 10: Segment Bisectors

## Benchmark

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.

## Learning Targets

- ☐ I can construct the bisector of a segment.
- ☐ I use the definition of a midpoint to solve problems.